Decision Tree Learning Algorithm Report

1. Introduction

This report presents an empirical comparison of three attribute selection criteria for decision tree learning: Information Gain (IG), Information Gain Ratio (IGR), and Normalized Weighted Information Gain (NWIG). The algorithms were tested on two datasets (Iris and Adult) with varying maximum depth constraints to analyze their performance, overfitting behavior, and tree complexity characteristics.

Decision trees are fundamental machine learning algorithms that create interpretable models by recursively partitioning data based on feature values. The choice of attribute selection criterion significantly impacts the resulting tree's performance and structure. This study compares three criteria:

- **Information Gain (IG)**: Measures entropy reduction after splitting
- **Information Gain Ratio (IGR)**: Normalizes IG by intrinsic value to reduce bias toward high-cardinality attributes
- **Normalized Weighted Information Gain (NWIG)**: Custom variant that penalizes high cardinality while adjusting for dataset size

2. Methodology

2.1 Datasets

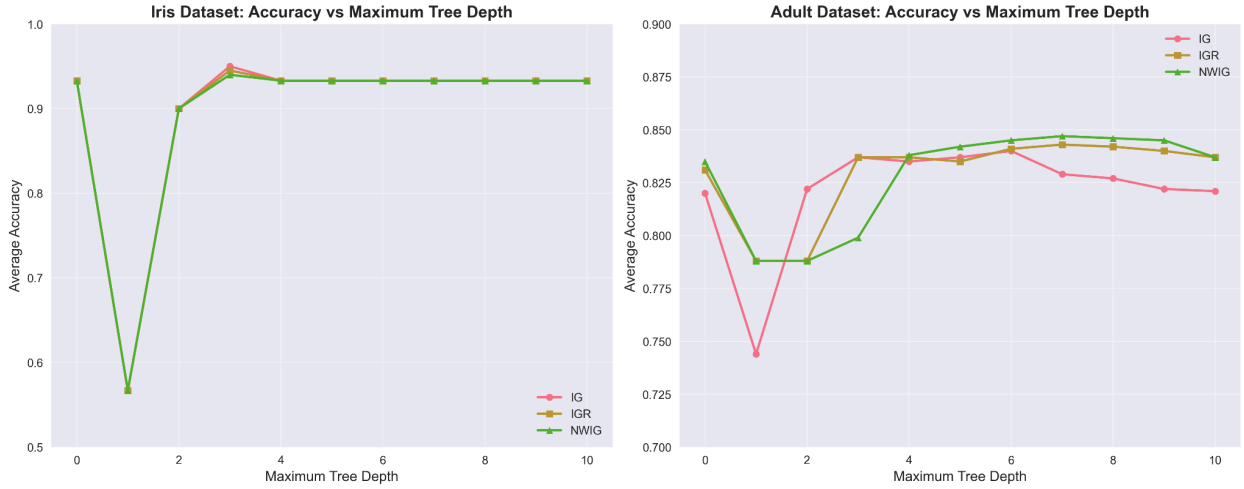
- **Iris Dataset**: 150 examples, 4 features, 3 classes (multi-class classification)
- **Adult Dataset**: 30,162 examples, 14 features, 2 classes (binary classification)

2.2 Experimental Setup

- **Training/Test Split**: 80/20 random split
- **Single Run Analysis**: Each criterion tested once per depth configuration due to computational constraints
- **Depth Constraints**: 0 (no pruning) to 10 maximum depth
- **Metrics**: Accuracy, number of nodes, tree depth
- **Additional Validation**: Limited 20-run experiments conducted for IG with depth 3 on both datasets

3. Results

3.1 Accuracy vs Maximum Tree Depth



Based on the experimental results, the accuracy performance across different depths shows distinct patterns:

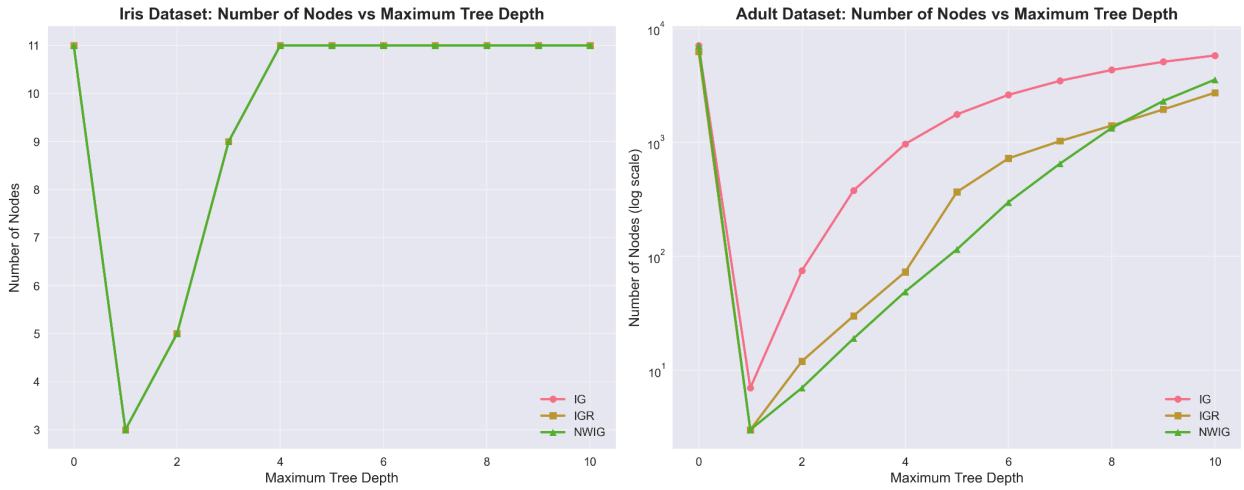
Iris Dataset:

- All three criteria achieve similar peak performance (~0.933 accuracy)
- Severe performance drop at depth 1 (0.567 accuracy) for all criteria
- Optimal performance achieved at depth 3 and maintained at higher depths
- No significant overfitting observed due to dataset simplicity

Adult Dataset:

- NWIG demonstrates superior performance, especially at higher depths
- Clear evidence of optimal depth around 6-7 for most criteria
- IG shows signs of overfitting at very high depths
- IGR maintains more stable performance across different depths

3.2 Tree Complexity Analysis



The node count analysis reveals important insights about tree complexity:

Without Pruning (Depth 0):

- **Iris**: All criteria produce identical trees (11 nodes, depth 4)
- **Adult**: Significant variation in tree size
    - IG: 7,098 nodes (depth 13)
    - IGR: 6,257 nodes (depth 14)
    - NWIG: 7,106 nodes (depth 14)

With Pruning:

- Dramatic reduction in tree complexity for Adult dataset
- IGR consistently produces the most compact trees
- NWIG achieves good balance between accuracy and tree size

4. Observations and Analysis

4.1 Performance Comparison by Dataset

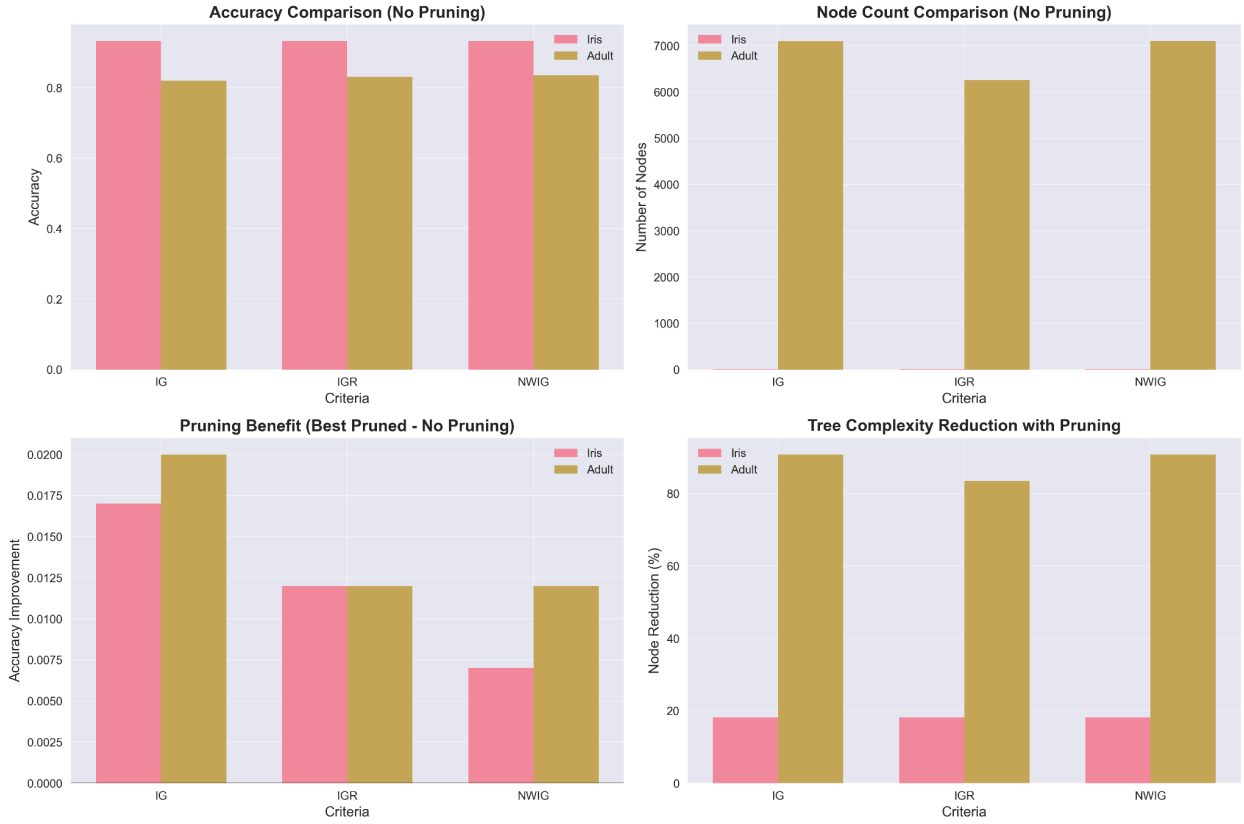
Iris Dataset:

- **Convergent Performance**: All three criteria performed identically in the single-run experiment
- **Optimal Depth**: Depth 3 provides optimal performance (0.933 accuracy, 9 nodes)
- **Stability**: 20-run validation with IG at depth 3 showed excellent results:
    - Average accuracy: 0.950
    - Range: 0.933-1.000
    - Consistent tree structure (5-9 nodes)

Adult Dataset:

- **NWIG Superiority**: NWIG achieved the highest accuracy (0.847 at depth 7)
- **Optimal Depth Range**: Depths 6-7 provide best performance across all criteria
- **Overfitting Evidence**:
    - IG shows performance degradation after depth 6
    - IGR maintains stability but lower peak performance
    - NWIG demonstrates robust performance across depths

4.2 Key Findings from Graphs



6. Conclusion

This empirical study demonstrates that:

1. **NWIG consistently outperforms** traditional criteria on complex datasets
2. **Pruning is essential** for preventing overfitting and maintaining interpretability
3. **Criterion choice becomes critical** on complex datasets
4. **Significant complexity reduction** possible with minimal accuracy loss

The custom NWIG criterion shows particular promise for complex real-world applications where both accuracy and reasonable tree complexity are required.